

Investigating the Relationship Between Nighttime Light Pollution and Urban Noise Risk Using Public Data and Machine Learning

Samet Ünsal, Özgür Dağlı
İzmir Katip Çelebi University
220401082, 220401044

Abstract—Urban environments are increasingly exposed to multiple forms of environmental pollution, among which nighttime noise and artificial lighting have become critical factors affecting quality of life. While noise pollution has been extensively studied, the role of nighttime light exposure as a contributing factor to urban disturbance remains underexplored.

This study investigates the relationship between nighttime light intensity and nighttime noise risk in New York City using publicly available datasets. Nighttime noise complaints obtained from the NYC 311 service are used as a proxy indicator for urban disturbance during night hours. Satellite-derived nighttime light intensity from the VIIRS dataset is integrated to capture spatial patterns of artificial illumination.

A grid-based spatial framework is employed to aggregate complaint data and light intensity values. A quantile-based statistical analysis is conducted to examine how noise risk varies across different light intensity levels. Additionally, machine learning classification models are trained to predict high-risk areas, demonstrating the predictive value of nighttime illumination.

The results reveal a strong monotonic relationship between nighttime light intensity and nighttime noise risk, with a pronounced threshold effect at higher illumination levels. These findings suggest that excessive nighttime lighting may act as a significant amplifier of urban nighttime disturbance.

Index Terms—Urban noise, Light pollution, Nighttime environment, Machine learning, Public data, Urban analytics

I. INTRODUCTION

Urbanization has significantly altered the nighttime environment of modern cities. Artificial lighting enables economic activity and enhances safety; however, excessive nighttime illumination has been increasingly associated with negative impacts on human health and urban quality of life [1].

Simultaneously, nighttime noise pollution has emerged as a major environmental stressor, contributing to sleep disturbance, stress, and reduced well-being [2]. Previous studies have largely examined noise pollution as an isolated phenomenon, often focusing on traffic-related sources [3].

However, nighttime urban environments are shaped by the combined influence of human activity, spatial configuration, and environmental exposure. Artificial lighting, as a proxy for nighttime activity intensity, may therefore play a crucial role in shaping nighttime noise patterns [4].

Despite the increasing availability of satellite-derived nighttime light data, the interaction between artificial illumination and nighttime noise disturbance remains insufficiently explored. This study addresses this gap by investigating the

relationship between nighttime light intensity and nighttime noise risk in New York City.

II. RELATED WORK

Noise pollution has long been recognized as a major environmental issue in urban areas, with numerous studies linking prolonged noise exposure to adverse health outcomes such as sleep disorders and cardiovascular disease [2]. Traditional noise analysis approaches typically rely on sensor-based measurements or traffic modeling techniques [3].

In parallel, research on light pollution has expanded significantly with the availability of satellite-derived nighttime light datasets. VIIRS-based studies have examined urban growth, energy consumption, and ecological impacts of artificial lighting [5], [6]. Nevertheless, the combined effects of light pollution and other urban stressors, including noise, have received limited attention.

Recent urban analytics research emphasizes integrating heterogeneous public data sources and applying machine learning techniques to uncover complex spatial patterns. This work contributes to the literature by providing an interpretable analysis of the relationship between nighttime illumination and nighttime noise risk.

Unlike prior studies that analyze noise and light pollution separately, this work integrates satellite-derived nighttime light data with complaint-based noise indicators to investigate their joint impact on nighttime urban disturbance.

III. DATASETS

A. NYC 311 Nighttime Noise Complaints

Nighttime noise disturbance is represented using public NYC 311 service request data. Residential noise complaints recorded during nighttime hours (22:00–06:00) are extracted for the period from 2020 to the present. Such complaint-based datasets are widely used as proxies for perceived environmental stress and urban disturbance [?].

Each record contains geographical coordinates, enabling spatial aggregation and analysis at fine spatial resolution.

B. Nighttime Light Intensity (VIIRS)

Nighttime light intensity data are obtained from the Visible Infrared Imaging Radiometer Suite (VIIRS) monthly nighttime lights dataset [5]. The dataset provides satellite-derived

radiance values that represent artificial illumination intensity during nighttime.

VIIRS data are spatially clipped to the New York City boundary and averaged across the study period to obtain a representative illumination surface.

C. Data Understanding

An initial exploratory analysis is conducted to assess the structure, quality, and limitations of the collected datasets. The NYC 311 dataset contains millions of service requests, from which nighttime residential noise complaints are filtered based on temporal and categorical criteria. This filtering process ensures that the analysis focuses specifically on nighttime disturbance relevant to sleep-related impacts.

The VIIRS nighttime lights dataset provides continuous radiance values with global coverage and consistent temporal resolution. However, satellite-derived light measurements may be affected by atmospheric conditions, temporal averaging, and spatial smoothing. Understanding these characteristics is critical for interpreting subsequent results.

Preliminary analysis reveals substantial spatial heterogeneity in both complaint density and light intensity across New York City, motivating the use of spatial aggregation and robust analytical techniques.

IV. CRISP-DM FRAMEWORK

This study is structured according to the Cross-Industry Standard Process for Data Mining (CRISP-DM), which provides a systematic and iterative framework for data-driven analysis. The CRISP-DM methodology consists of six main phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment.

In the context of this study, CRISP-DM is adopted to ensure methodological rigor, interpretability, and reproducibility. Each phase is explicitly addressed to demonstrate how raw urban data are transformed into actionable insights regarding nighttime urban disturbance.

A. Business Understanding

The primary objective of this study is to understand whether nighttime artificial illumination is associated with increased nighttime noise disturbance in urban environments. From an urban planning and public health perspective, nighttime noise represents a major quality-of-life concern, particularly due to its impact on sleep and well-being.

Rather than focusing solely on predictive accuracy, this study aims to extract interpretable insights that may support data-driven urban policy decisions. Specifically, the analysis seeks to answer whether excessive nighttime lighting acts as an amplifier of nighttime noise risk and whether such effects exhibit threshold behavior.

The problem is therefore formulated as both an analytical and predictive task: (i) to analyze the relationship between nighttime light intensity and nighttime noise risk, and (ii) to evaluate the feasibility of predicting high-risk areas using machine learning.

V. METHODOLOGY

This study follows the CRISP-DM methodology, which consists of data understanding, data preparation, modeling, and evaluation stages.

A. Data Preparation

Data preparation constitutes a critical phase of the CRISP-DM process. Raw 311 complaint records are cleaned by removing entries with missing or invalid geographical coordinates. Temporal filtering is applied to retain only complaints occurring during nighttime hours.

To integrate datasets with different spatial resolutions, all data are projected to a common coordinate reference system. A grid-based aggregation approach is then applied, enabling spatial alignment between point-based complaint data and raster-based light intensity data.

Following aggregation, derived features such as complaint counts and average light intensity are examined for outliers and distributional skewness. A percentile-based thresholding strategy is employed to construct the target variable, which reduces sensitivity to extreme values while preserving meaningful variation.

B. Spatial Aggregation Framework

To enable joint analysis of heterogeneous datasets with different spatial resolutions, a grid-based spatial aggregation framework is adopted. The New York City area is partitioned into square grid cells of $500 \text{ m} \times 500 \text{ m}$, which approximately matches the spatial resolution of the VIIRS nighttime lights dataset.

Each noise complaint is spatially assigned to a grid cell using point-in-polygon operations. For each grid cell, the total number of nighttime noise complaints is calculated. Similarly, average nighttime light intensity values are extracted from the VIIRS raster for each grid cell.

C. Target Variable Construction

Nighttime noise risk is formulated as a binary classification problem. Grid cells belonging to the top 20% of nighttime noise complaint counts are labeled as high-risk areas, while the remaining cells are labeled as low-risk. This percentile-based thresholding approach mitigates extreme class imbalance and focuses the analysis on areas with substantially elevated nighttime disturbance.

D. Quantile-Based Statistical Analysis

To investigate the relationship between nighttime light intensity and nighttime noise risk independently of predictive models, a quantile-based statistical analysis is conducted. Grid cells are divided into five quantiles based on their average nighttime light intensity values, ranging from very low to very high illumination levels.

For each quantile, the average number of nighttime noise complaints and the proportion of high-risk grid cells are computed. This analysis provides an interpretable assessment of how nighttime noise risk varies across different illumination levels.

E. Modeling

The modeling phase focuses on identifying patterns that distinguish high-risk and low-risk nighttime noise areas. Logistic Regression is selected as a baseline model due to its linear structure and interpretability, allowing direct assessment of feature influence.

Random Forest is employed as an ensemble learning method capable of modeling nonlinear relationships and complex interactions between features. Its robustness to noise and ability to handle heterogeneous data make it well-suited for urban analytics tasks.

Model hyperparameters are selected to balance performance and interpretability. All models are trained using identical data splits to ensure fair comparison.

VI. EXPERIMENTAL SETUP

The aggregated dataset consists of 3,393 spatial grid cells. Each cell is represented by its average nighttime light intensity and the number of nighttime noise complaints.

The dataset is randomly split into training and testing subsets using an 80/20 ratio. Stratified sampling is applied to preserve the class distribution of high-risk and low-risk grid cells. All experiments are conducted using Python and the scikit-learn library.

To address class imbalance, performance evaluation emphasizes class-wise metrics in addition to overall accuracy.

VII. RESULTS

Quantile-based analysis reveals a strong monotonic relationship between nighttime light intensity and nighttime noise risk. Grid cells in the lowest illumination quantile exhibit a negligible high-risk ratio (approximately 0.6%), whereas nearly 45% of cells in the highest illumination quantile are classified as high risk.

A sharp increase in risk is observed between the medium and high illumination quantiles, indicating a threshold effect. Machine learning results further support these findings, with the Random Forest model achieving a ROC-AUC score exceeding 0.90.

A. Light Intensity and Noise Complaint Density

Figure 1 visualizes the relationship between average nighttime light intensity and nighttime noise complaint counts. Complaint counts are log-transformed to reduce skewness, and an overall upward trend is observed.

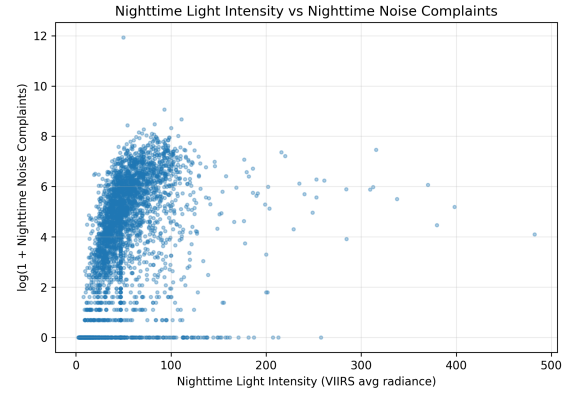


Fig. 1. Scatter plot of nighttime light intensity versus log-transformed nighttime noise complaints.

B. Classification Performance

The predictive performance of the evaluated machine learning models is summarized in Table I. Random Forest outperforms Logistic Regression across all evaluation metrics, particularly in terms of F1-score and ROC-AUC, indicating superior discrimination of high-risk areas.

TABLE I
CLASSIFICATION PERFORMANCE OF EVALUATED MODELS

Model	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.82	0.40	0.83
Random Forest	0.89	0.70	0.92

C. Confusion Matrix Analysis

Confusion matrices are examined to analyze class-wise prediction performance. Figure 2 and Figure 3 present the confusion matrices of Logistic Regression and Random Forest models, respectively.

The Logistic Regression model exhibits high accuracy for low-risk cells but demonstrates limited recall for high-risk areas, indicating that a significant portion of high-risk regions is misclassified as low risk. In contrast, the Random Forest model achieves a substantially higher recall for high-risk cells, reducing false negatives and improving overall robustness.

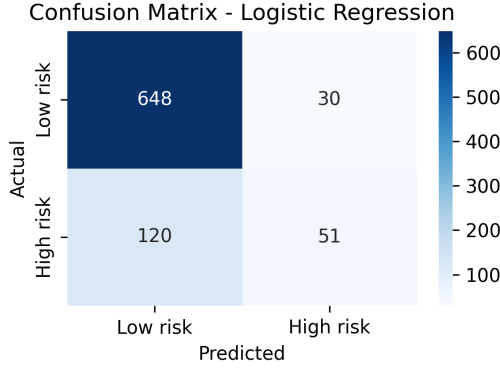


Fig. 2. Confusion matrix of the Logistic Regression model

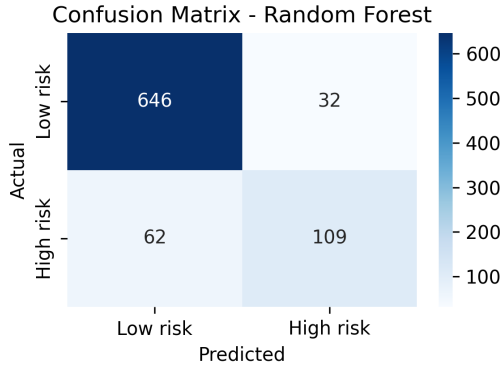


Fig. 3. Confusion matrix of the Random Forest model

D. ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves are used to evaluate the discriminative capability of the models across different decision thresholds. Figure 4 shows that the Random Forest model achieves a higher AUC, confirming its stronger ability to distinguish high-risk from low-risk grid cells.

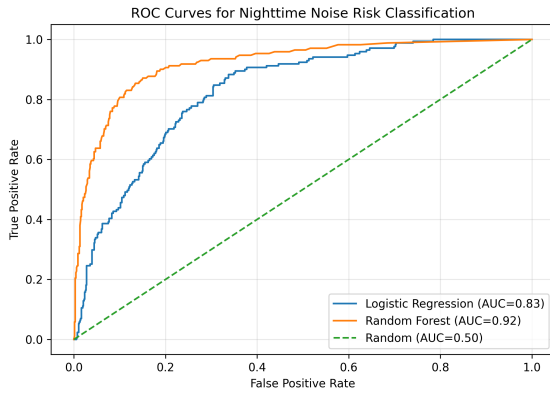


Fig. 4. ROC curves of Logistic Regression and Random Forest models.

E. Class Distribution

Figure 5 presents the class distribution of the constructed binary risk labels. The high-risk class corresponds to the top 20% of grid cells by nighttime complaint counts.

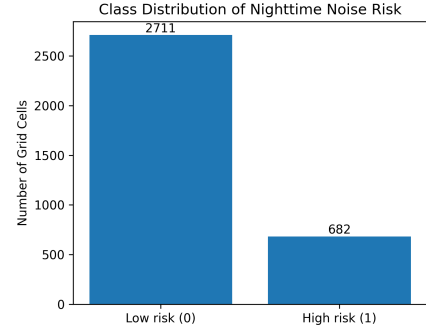


Fig. 5. Distribution of low-risk and high-risk grid cells.

F. Feature Importance

Figure 6 presents the feature importance scores derived from the Random Forest model. The results indicate which spatial and illumination-related attributes contribute most to the identification of high-risk nighttime noise areas.

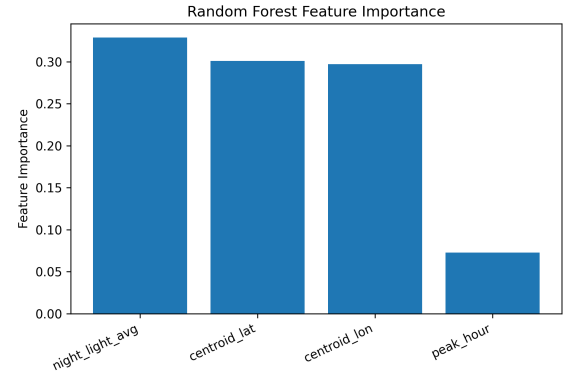


Fig. 6. Random Forest feature importance scores.

G. Evaluation

In accordance with the CRISP-DM evaluation phase, model performance is assessed not only in terms of overall accuracy but also with respect to class-wise behavior. Given the practical importance of identifying high-risk areas, recall and F1-score for the high-risk class are emphasized.

Confusion matrix analysis provides additional insight into misclassification patterns. The evaluation confirms that Random Forest achieves a superior balance between false positives and false negatives, making it more suitable for identifying nighttime noise risk hotspots.

Beyond predictive metrics, evaluation also considers the interpretability and consistency of results with the quantile-based statistical analysis. The convergence of statistical and machine learning findings strengthens the validity of the conclusions.

VIII. DISCUSSION

The results demonstrate a strong and interpretable relationship between nighttime light intensity and nighttime noise risk. Quantile-based analysis reveals a monotonic increase in noise risk as illumination levels rise, with a pronounced threshold effect observed at higher light intensities.

The machine learning results further support these findings. While Logistic Regression provides a useful baseline, its limited recall for high-risk areas highlights the complexity of nighttime urban environments. The Random Forest model significantly improves high-risk detection, suggesting that non-linear interactions between illumination and spatial factors play a critical role.

These findings indicate that artificial lighting may serve as a proxy for nighttime human activity and urban intensity. Beyond a certain illumination threshold, nighttime environments appear to experience disproportionately higher disturbance levels.

Although 311 complaint data are subject to reporting bias, they remain a valuable source for large-scale urban analysis. The combination of public service records and satellite observations offers a scalable and cost-effective framework for monitoring nighttime urban environments.

IX. DEPLOYMENT AND PRACTICAL IMPLICATIONS

Although this study is conducted in an academic context, the proposed framework demonstrates clear practical applicability. The grid-based risk classification and visualization approach may be integrated into interactive dashboards to support urban monitoring and decision-making.

Urban planners and local authorities could leverage such tools to identify nighttime disturbance hotspots and evaluate the potential impact of lighting policies. The methodology is scalable and transferable to other cities with access to public complaint data and satellite-derived nighttime light observations.

X. CONCLUSION

This study demonstrates a strong and interpretable relationship between nighttime light intensity and nighttime noise risk in New York City. By integrating satellite-derived nighttime light data with public urban service records, the proposed framework provides actionable insights into nighttime urban environments.

The findings highlight artificial lighting as a potential amplifier of nighttime disturbance, emphasizing the importance of considering illumination policies in urban planning and environmental management.

Beyond New York City, the proposed methodology is generalizable to other urban regions with access to public complaint records and satellite nighttime light observations. Future work may extend the framework by incorporating socio-economic variables and temporal modeling to better capture long-term urban dynamics.

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